

Nguyễn Mạnh Tú

Software Engineer - AI Engineer

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SUMMARY

Backend Software Engineer with 2+ years of experience building REST APIs, batch jobs, and asynchronous backend services using Java, Spring Boot, PostgreSQL, AWS SQS, Docker, and LocalStack. Experienced in backend feature development, database design, code review, sprint execution, and supporting a 4-5 member sub-team. Recently focused on AI application engineering, including semantic search, RAG systems, CV/JD matching, and AI interview platforms using FastAPI, Qdrant, OpenAI, LiveKit, and cloud storage.

EXPERIENCE

Software Engineer / Backend Developer

5/2024 - 3/2026

FPT Software

Developed and maintained backend APIs for recruitment and HR-related systems using Java, Spring Boot, and PostgreSQL, resulting in a 20% increase in system efficiency.

- Implemented batch jobs and asynchronous processing flows using AWS SQS and backend scheduling mechanisms.
- Worked with Docker and LocalStack to support local development and testing of cloud-integrated services.
- Reviewed code, supported teammates, and contributed to sprint execution as a sub team lead for a 4-5 member backend sub-team.
- Coordinated task assignments, tracked progress, and performed technical reviews, resulting in a 30% boost in team productivity during the main team lead's absence.
- Developed REST APIs for recruitment and HR-related systems, ensuring seamless service integration and data flow.

PROJECTS

Financial Planning System

Expense management and tracking system, supporting enterprises in transparentizing financial data and internal processes.

Developed the backend with Java Spring Boot, building RESTful APIs for expense management and reporting functions.

Built the frontend interface with React and Tailwind CSS.

Applied Redis for caching and improving query performance.

Java · Spring Boot · React · SQL Server · Redis

github.com/tuannqhe151337/fin-planning-backend.git

Scholar AI — AI-Augmented Research Workspace

Led the development of an AI-native research workspace that helps users discover, read, synthesize, and write from academic papers.

Designed a RAG-based workflow with hybrid retrieval using dense embeddings and BM25-style keyword matching to improve both semantic and exact-term search.

Implemented citation-aware chat to ground answers in retrieved paper context and reduce hallucination risk.

Designed hierarchical chunking with parent-child mapping to retrieve precise passages while preserving enough context for LLM answers.

Added guardrails such as cosine similarity thresholding and scope-enforcing prompts to prevent off-topic or unsupported answers.

Built an evaluation workflow using LangSmith with metrics such as retrieval hit rate, citation accuracy, answer presence, refusal correctness, correctness, and faithfulness.

Presented the project through a live demo and technical pitch as part of an AI practical training program.

Next.js · React · FastAPI · PostgreSQL · Qdrant Cloud · OpenAI · Supabase · Render · Vercel · LangSmith

github.com/TuNM17421/ScholarAI

VCareer — AI Career Development Platform

Contributed to the development of VCareer, an end-to-end AI career development platform covering real-time AI interviews, CV improvement, CV/JD matching, portfolio generation, job applications, recruiter workflows, and career support.

Designed and implemented features using a Next.js full-stack architecture with Server Components, Server Actions, API Route Handlers, Zod validation, and Drizzle ORM over Neon PostgreSQL.

Built and integrated the AI Interview module with real-time voice interview flows using WebRTC, LiveKit, OpenAI Realtime, and Gemini Live as alternative AI engines.

Implemented interview session handling, turn logging, recording storage, and asynchronous scoring workflows with Cloudflare R2 and queue-based evaluation.

Integrated a 3D TalkingHead avatar layer using three.js to support more realistic AI interviewer interactions.

Developed CV Suite features including CV builder, PDF rendering, PDF/DOCX parsing, base CV quality check, CV review, and CV-JD matching workflows.

Designed CV analysis flows to identify missing information, ATS-readiness issues, JD alignment gaps, and improvement suggestions based on structured evaluation evidence.

Worked on Jobs & Applications workflows including job posts, job questions, saved jobs, application tracking, and JD crawler/adaptor logic for external job sources.

Next.js 16 · React 19 · TypeScript · Tailwind CSS v4 · shadcn/ui · Drizzle ORM · Neon PostgreSQL · Better Auth · Cloudflare R2 · Upstash Redis · OpenAI Realtime/GPT · Google Gemini Live · LiveKit · Zod · Vercel

github.com/TuNM17421/VCareer

EDUCATION

FPT University

Bachelor’s Degree, Information Technology

GPA: 7.62 / 10

2019 – 2024

SKILLS

• Java	• Spring Boot	• REST APIs
• Batch processing	• Maven/Gradle	• Git
• SQL Server	• SQL	• AWS SQS
• Docker	• LocalStack	• Cloudflare R2
• JavaScript	• TypeScript	• React
• Render	• LLM	• RAG
• Prompt Engineering	• Qdrant	• FastAPI

- LangSmith
 - Code Review
- LiveKit
 - Sprint Tracking
- Next.js
 - Team Coordination

CERTIFICATIONS

- TOPIK II – Level 3

LANGUAGES

English (Independent) · Korean (TOPIK II – Level 3)